Problem Solving and Calculations 3

Set 3: Circular Motion

Notes

1. An Olympic hammer thrower whirls the hammer, which is a round metal ball on the end of a short steel wire, rapidly in a circle in preparation for the throw. If the athlete wants the hammer to travel due west, at what point should the athlete release the hammer? A diagram may help when you explain your answer.

2. Why does a sprinter running in a 200 m event lean towards the centre of the curve he is rounding?

3. A roller skater coasts around a curve at constant speed on a horizontal surface. What provides the centripetal force?

4. Use a diagram to clearly explain why engineers design banked curves on roads that have high speed limits.

5. An ice skater glides around a curve of radius 15 m at a constant speed of 3.5 m s^{-1}. What is the skater's acceleration?

6. A baseball player swings a 0.585 kg bat in a horizontal arc so that its centre of mass moves in a curved path of radius 1.25 m at a constant speed of 11.5 m s^{-1}. Determine the magnitude of the force with which the player must grip the bat.

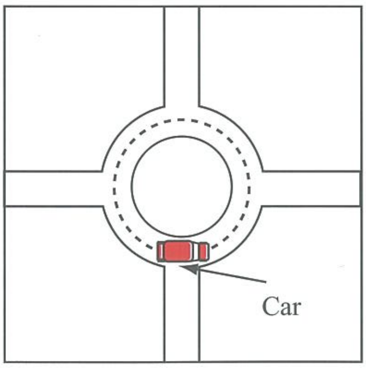
7. A playground roundabout takes 15.5 s to make a complete revolution.
   a) Calculate the speed of a child sitting on the roundabout 3.80 m from its centre.
   b) Calculate the centripetal force on the child if her mass is 28.0 kg.

8. A civil engineer has to design a road in which there is a curve with a radius of 300 m. The road will have a maximum speed limit of 110 km h^{-1}. At what angle should the road bank on the curve so that no frictional force is needed by a vehicle, travelling at the speed limit, to move round it?

9. Susanna swings on a 4.00 m long maypole chain with enough speed to swing in a circle of radius 2.50 m. She has a mass of 55.0 kg.
   a) Find the tension in the chain.
   b) Find her period of revolution.

Question 9

10. A 1250 kg car follows a circular path around a roundabout of radius 18.0 m at a constant speed of 24.0 km h^{-1}.
    a) Is the car accelerating? Explain.
    b) Find the average force provided by the friction between the tyres and the road to maintain this circular path.
    c) At what angle would the curved road need to be banked for there to be no need to rely on friction to maintain the circular path?



Question 10

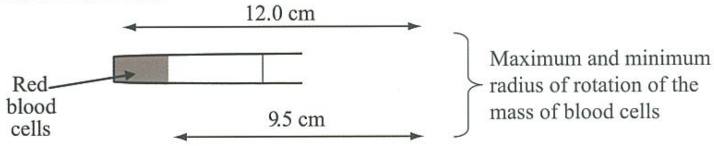
Problem Solving and Calculations

Set 3: Circular Motion

Notes

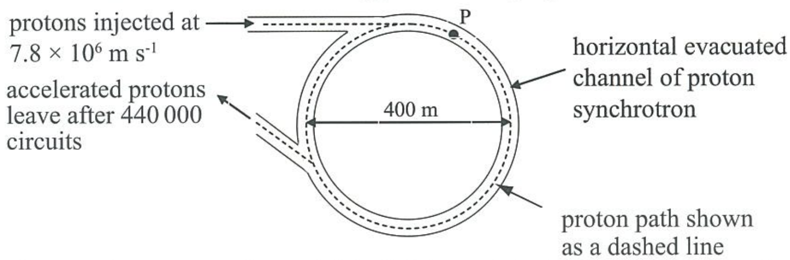
11. The track at a velodrome is banked at an angle of 20°. The radius of the track is 70.0 m. Calculate the minimum speed that a cyclist must maintain in order to stay on the track without relying on friction.

12. Red blood cells are separated from plasma using a centrifuge. Little test-tubes of blood are loaded into the centrifuge and then rotated at high speed. The test tubes swing outwards and the red blood cells move into the ends of the test-tubes. The plasma is poured off, leaving the red blood cells behind.

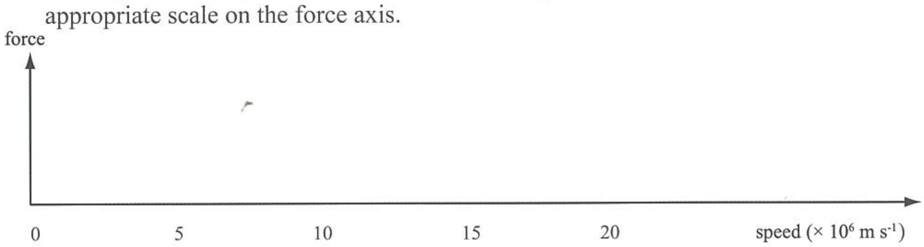


a) If the centrifuge in the diagram is rotating at 3800 revolutions per minute, determine the minimum speed of the red blood cells in the test tubes.
b) Calculate the maximum centripetal acceleration of the red blood cells in the test tubes.
c) A red blood cell has a mass of approximately 98 ng. Red blood cells break if subject to a force greater than 8.2 mN. Determine the maximum rotational frequency of this centrifuge if the cells are not to be damaged.

13. The diagram below shows the schematic diagram (as viewed from above) of a proton synchrotron. This is a device for accelerating protons to high speeds in a horizontal circular path.



In the synchrotron the protons of mass 1.7×10^{-27} kg are injected, as shown in the diagram, at a speed of 7.8×10^6 m s^{-1}. The diameter of the path taken by these protons is 400 m.
a) Show on the diagram the direction of the force required to make a proton move in the circular path when it is at the position marked P.
b) Calculate the force that has to be provided to produce this path for this proton.
c) Sketch, on the grid below, a graph that shows how this force will have to change as the speed of the proton increases over the range indicated on the x-axis. Include an appropriate scale on the force axis.



Before reaching their final energy the protons in the synchrotron travel around the accelerator 440 000 times in 2.5 s.
d) Calculate the total distance travelled by a proton in the 2.0 s time interval.
e) What would happen to the vertical displacement of the proton in this time?
f) Consider your answer to e) above; what must be added to the synchrotron?

Problem Solving and Calculations 3

Set 3: Circular Motion

14. A string just supports a hanging brick without breaking. Explain why the string breaks if you set the brick swinging.

Notes

15. (a) Estimate the minimum speed required to spin a bucket of water at arm's length in a vertical loop without spilling the water.
    (b) Explain why the water does not fall out if the bucket traverses the top of its path at this or greater speed.
    (c) Does the bucket travel at a constant speed throughout its circular path? Explain.

16. A pilot flies her aeroplane in a vertical loop of diameter 1.60 km.
    (a) How fast is the aeroplane travelling at the top of the loop if the pilot feels no force from either the seat or the straps?
    The pilot cuts the engine at the top of the loop.
    (b) Ignoring air resistance, what is the speed of the aeroplane as it emerges from the bottom of the loop?

17. An aeroplane flies in a vertical loop of radius 650 m. At the top of the loop, the pilot experiences a downward reaction force, from her the seat, equal to one fifth of her weight. Calculate the aeroplane's speed at this instant.

18. A model car of mass 2.00 kg moves in a vertical circle of radius 5.00 m. If its speed at the lowest point is 20.0 m s^{-1} and at the highest is 10.0 m s^{-1}, calculate
    (a) the force that the track exerts on the car at the lowest point;
    (b) the force that the track exerts on the car at the highest point.

19. You strap into a safety harness and take a roller coaster ride. In one part of the ride, the roller coaster car goes through a vertical loop at a speed of 14.0 m s^{-1}.
    (a) Calculate the radius of the loop of track if you feel "weightless" as you pass through the top of the loop.
    (b) Describe what would happen to you if the car went through the loop faster than 14.0 m s^{-1}. Explain your answer.
    (c) Describe what would happen to you if the car went through the loop slower than 14.0 m s^{-1}. Explain your answer.

20. As a 40.0 kg gymnast swings in a vertical circle on a high bar, her centre of mass moves around 0.90 m from the bar.
    (a) At the highest point her centre of mass is moving at 1.00 m s^{-1}. Sketch a free body diagram for this situation.
    (b) How fast is she moving when her centre of mass is level with the bar? Sketch a free body diagram for this situation.
    (c) How much force must she exert on the bar in order to hang on as she passes through the lowest point of her swing? Sketch a free body diagram for this situation.

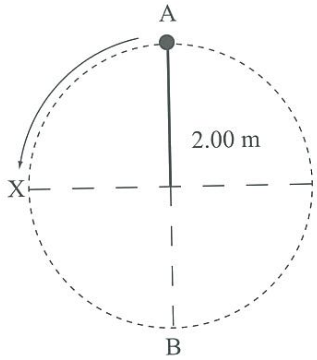
Problem Solving and Calculations

Set 3: Circular Motion

Notes

21. Passengers on a fairground ride revolve at a constant speed in a vertical circle of radius 3.60 m. The ride operator has a choice of two speeds, LOW and HIGH. At the HIGH setting, passengers feel weightless at the top of the circle; at the LOW setting, the passengers revolve at half the HIGH speed.
    (a) Draw free body diagrams showing the forces acting on a passenger at the top and at the bottom, at each speed setting. (That's four diagrams altogether.)
    (b) Calculate the speed at which the ride moves, at the HIGH setting.
    (c) Calculate the reaction forces acting on a passenger of mass 60.0 kg at the top and bottom of the circle, when travelling at the HIGH setting.
    (d) Calculate the reaction forces acting on a passenger of mass 60.0 kg at the top and bottom of the circle, when travelling at the LOW setting.

22. A stone of mass 2.50 kg is whirled in a vertical circle at the end of a 2.00 m length of string.



(a) The stone passes through point X at a speed of 10.4 m s^{-1}. Calculate its speed at points A and B.
(b) Calculate the tension in the string at points A and B.
(c) At which point, A, B or X, is the string most likely to break? Explain your answer.

Investigation 3.2: Centripetal forces in action

3

Notes

Background

There are a number of everyday situations where centripetal forces producing horizontal circular paths are evident – cars turning a corner, athletes running the bend of a race track, high speed trains turning on banked tracks, aircraft banking, etc.

The task

Use diagrams and appropriate mathematics to explain the origin of the centripetal force in three different situations (i.e. do not do an athlete running around a track and a cyclist negotiating a race track, as these are predominantly the same example).

Motion & Forces in a Gravitational Field

Set 1	2 a)	18 m N 56° E
	2 b)	22 m N 63° W
	3	294 N
	4	3.8 m s^{-1} at 23° to rip
	5 a)	42°
	5 b)	3.4 m s^{-1} at 53° to the bank
	5 c)	30 m
	6 a)	9.2 km
	6 b)	N 41° W
	7	5.5 m s^{-1} away from the player
	8	47 m s^{-1} toward the opponent
	9	32 m s^{-1} at 51° to the final velocity
	10	212 N in the forward direction
	11	42 m s^{-1} at 45° to both initial and final velocities
	12 a)	Position 1: 427 N toward the Earth
	12 b)	Position 2: 359 N at 2.10° to the line joining the Earth and the asteroid
	13	43 m s^{-1}
	14	1.78 m s^{-1} N 38.2° E
	15 a)	1.17×10^3 N
	15 b)	125 N
	16	14 m s^{-1}, 20 m s^{-1}
	17	4.14 m s^{-2} down the slope
	18	2.5 s
	19	177 N perpendicular to, and toward, the path of the boat
	20 a)	3.5 s
	20 b)	30 m
	21 b)	32.6°
	22	716 N at 2.7° to the left of the boat's path
Set 2	6	5 points
	7	3.48 m
	9	1.04 s
	10	1.16 m
	11 a)	2.1 s
	11 b)	31 m
	12	78.8 m
	13	0.81 m
	14	285 m
	15 a)	vertical = 9.0 m s^{-1} upward; horizontal = 9.1 m s^{-1}
	15 b)	6.5 m
	15 c)	18.9 m
	16 a)	vertical = 29.0 m s^{-1}; horizontal = 7.76 m s^{-1}
	16 b)	No
	16 c)	29.0 m s^{-1} to the right
	16 d)	Assuming she hits the ramp with her foot already fully down, then at point A; Ignoring air resistance, the speed at point A should equal the speed at point E
	16 e)	27.3 m s^{-1}
	17 c)	Yes, by 9.2 m
	18	35.6 m
	19	8.10 m s^{-1}
	20 a)	no
	20 b)	yes

Set 3	5	0.82 m s^{-2} toward the centre of the circle
	6	61.9 N
	7 a)	1.54 m s^{-2}
	7 b)	17.5 N
	8	17.6°
	9 a)	691 N
	9 b)	3.55 s
	10 a)	yes
	10 b)	3.09 kN
	10 c)	15.3°
	11	15.8 m s^{-1}
	12 a)	37.8 m s^{-1}
	12 b)	1.90×10^{4} m s^{-2}
	12 c)	17.7 kHz
	13 b)	2.59×10^{-16} N
	13 d)	5.53×10^{8} m
	13 e)	it would drop 30.6 m
	16 a)	88.5 m s^{-1}
	b)	198 m s^{-1}
	17	87.4 m s^{-1}
	18 a)	180 N upward (b) 20.4 N downward
	19 a)	20 m
	20 b)	4.32 m s^{-1}
	c)	2.00×10^{3} N
	21 b)	5.94 m s^{-1}
	c)	at top, 0; at bottom, 1176 N upward
	d)	at top, 441 N upward; at bottom, 735 N upward
	22 a)	at A: 8.30 m s^{-1}; at B: 12.1 m s^{-1}
	b)	at A: 61.6 N; at B: 208 N
Set 4	6	6×10^{24} kg
	7	1.72×10^{-6} N
	8 a)	2.64×10^{6} m
	8 b)	8.16 m s^{-2} toward the Earth
	8 c)	7.55×10^{3} m s^{-1}
	9	3.80×10^{8} m
	11 b)	2.38×10^{20} N toward the Sun
	18	5.74×10^{3} s (1.59 hours)
	19 a)	1.37×10^{4} m s^{-1}
	19 b)	1.90×10^{27} kg
	20	5.97×10^{24} kg
	21	3.59×10^{7} m
	22	3.05 × (radius of Moon's orbit around Earth)
	23 a)	2.07×10^{22} N; 9.20×10^{21} N
	23 b)	5.37×10^{4} m s^{-1}; 4.39×10^{4} m s^{-1}
Set 5	2	120 N m
	3	220 N
	6 b)	18 kg
	8 a)	94 N
	8 b)	447 N
	9	46 kg
	10 a)	630 N
	10 b)	0.75 m toward Q
	11	1.5 m from the front wheels
	12 a)	18 kN
	12 b)	0.375 × length of log from the heavier end